

HydroChirp: Dynamic Chirp Shaping for Reliable and Ultra-Long-Range Underwater Communication

Chiyu Zhou*, Zijian Wan*, Baodong Chen[†], Zicheng Chi[‡], Xin Liu[§], Wei Wang*

* Department of Computer Science, Texas Tech University, {chizhou, ziwan, wei.wang}@ttu.edu

[†] Department of Computer Science and Engineering, The Ohio State University, chen.12610@osu.edu

[‡] Department of Electrical Engineering and Computer Science, Cleveland State University, z.chi@csuohio.edu

[§] Department of Computer Science, Florida State University, xliu15@fsu.edu

Abstract—Recent advances in underwater communication techniques have shown significant potential for supporting diverse applications. However, little work has been done to achieve reliable and ultra-long-range underwater communication using commodity mobile devices in various underwater environments. In this paper, we present *HydroChirp*, a novel acoustic-signal-based underwater communication system designed to achieve reliable and ultra-long-range communication using commodity mobile devices. To do this, *HydroChirp* dynamically selects the channels with minimal distortion and leverages a novel chirp segmentation technique to mitigate the impact of generic and low-bandwidth noises. To address frequency offsets, *HydroChirp* uses an innovative frequency-noise model to estimate and compensate for noise generated by water fluctuations and determine the optimal transmission power. Furthermore, *HydroChirp* enhances communication reliability and range by analyzing noise patterns generated by specific underwater objects. We implement *HydroChirp* on commodity mobile devices and extensively evaluate its performance across various underwater scenarios. The experimental results demonstrate that *HydroChirp* can achieve a communication range of up to 150 meters with high reliability and efficiency.

Index Terms—Underwater Communication, Mobile Sensing, Internet of Things.

I. INTRODUCTION

Approximately two-thirds of the Earth’s surface is covered with water. Due to the increasing need for effective monitoring, exploration, and connectivity in aquatic environments, underwater communication has become one of the most famous research topics recently [1], [2]. Beyond industrial and scientific applications, underwater communication techniques have also shown great potential to enhance personal recreational activities, such as scuba diving [3], [4], underwater exploration [5], [6], and snorkeling [7], [8]. Meanwhile, the rapid proliferation of mobile devices (e.g., smartphones) equipped with built-in sensors offers a promising opportunity to develop cost-effective and widely accessible underwater communication solutions. These devices have the potential to bridge the gap between advanced underwater communication technologies and everyday users.

To enable underwater communication, one of the most famous solutions is to use optical signals [9]–[11]. For example, laser-based techniques have been employed to support underwater communication [9], [10]. However, these solutions often rely on expensive hardware and are constrained by limited communication ranges [11], which restricts their potential in

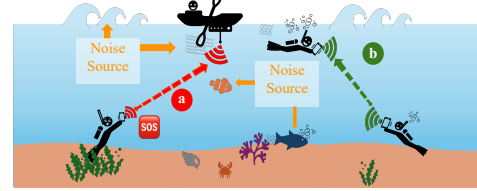


Fig. 1: An example of the applications of *HydroChirp* in noisy underwater environments: (a) transmitting SOS messages, and (b) supporting personal recreational activities.

many underwater scenarios. To overcome these limitations, acoustic signals have emerged as a viable option for underwater communication. For instance, AquaApp [7] introduced a high-throughput underwater communication system by using OFDM-based acoustic signals. Recently, AquaHelper [12] can use cheap acoustic receivers to detect weak acoustic SOS signals in underwater scenarios.

Although existing solutions perform well in their application scenarios, little work has focused on achieving reliable and long-range underwater communication using commodity mobile devices. In this paper, we introduce *HydroChirp*, a novel acoustic-signal-based system for reliable and ultra-long-range communication in diverse underwater environments. Figure 1 illustrates an example of the potential applications for *HydroChirp*. In (a), *HydroChirp* enables the transmission of SOS messages using a mobile phone in various underwater scenarios, regardless of noise levels. In (b), *HydroChirp* supports reliable and ultra-long-range underwater communication for personal recreational activities (e.g., scuba diving). To build *HydroChirp*, we leverage carefully-designed chirp signals generated by the speakers in commodity mobile devices (i.e., smartphones) and use Chirp Spread Spectrum (CSS) [13] for underwater communication. This is because chirps are more resistant to channel noise and have the potential to enable long-range communication. However, we still need to answer the following questions:

I. How to dynamically select channels with relatively low noise and attenuations? In underwater environments, noise is *unevenly distributed* across the spectrum. Studies have shown that certain frequency bands may experience higher levels of generic acoustic noise and attenuation than others, which significantly reduces communication reliability [7], [14]. Therefore, to conduct reliable and ultra-long-range

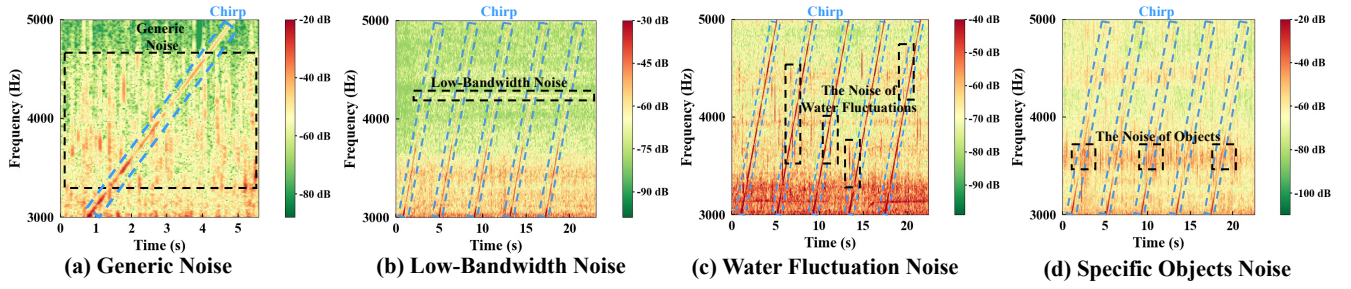


Fig. 2: The Spectrum of the Noise in Underwater Environments

underwater communication, it is important to dynamically select low-noise channels. However, this task is particularly challenging. *First*, since the size of the speakers in commodity mobile devices (e.g., smartphones) is limited, they cannot generate high-frequency acoustic signals with good quality and enough power [15]. As a result, although higher frequency bands often exhibit lower noise levels, mobile devices must rely on mid-frequency bands for underwater communication. *Second*, based on our observation, due to uncertainties in the underwater environment, even frequency bands with low generic noise levels can be affected by some low-bandwidth noises. These low-bandwidth noises will introduce high distortions to the transmitted acoustic signals. To overcome these challenges, *HydroChirp* introduces a novel **Dynamic Sub-Channel Selection** module that dynamically selects low-distortion sub-channels and segments chirp signals to mitigate generic noise and avoid low-bandwidth interference.

II. How to mitigate interference generated by water fluctuations and determine the optimal transmission power? In practice, due to the water fluctuations, the surface and internal waves can introduce significant frequency offset and noise to the communication channel [16], [17]. Moreover, the channel characteristics (e.g., noise and attenuation) vary across different scenarios. For example, the noise level differs significantly between a calm lake and a turbulent river. In addition, as the communication distance increases, the channel becomes increasingly random and inconsistent due to higher attenuation and interference introduced by water fluctuations. To address these challenges, *HydroChirp* introduces a novel **Hydrodynamic Estimation and Compensation** module, which leverages frequency offsets to estimate noise levels caused by water waves. Then, based on the estimated results, *HydroChirp* can determine the optimal transmission power to enable reliable and ultra-long-range underwater communication.

III. How to mitigate noise and interference generated by specific objects? There are many ambient objects that can impact the underwater acoustic communication. For example, raindrops, swimming fish, and paddling activities can generate noise and contribute to interference, which varies dynamically due to the motion of these objects [18], [19]. To overcome this challenge, *HydroChirp* introduces a novel **Sensing-Based Signal Enhancement** module to sense and estimate the noises generated by specific underwater objects. Then, to improve communication reliability, it leverages a dynamic spreading factor (SF) adjustment scheme to adjust the divided chirp

segments. At last, the receiver demodulates the acoustic signals by using a dynamic demodulation strategy.

We summarize the contributions of *HydroChirp* as follows:

- To our knowledge, this is the first work that focuses on reliable and ultra-long-range underwater communication using commodity devices. We believe *HydroChirp* has the potential to make underwater communication practical to general users.
- We introduce multiple innovative techniques to address the challenges of underwater communication, including Dynamic Sub-Channel Selection, Hydrodynamic Estimation and Compensation, and Sensing-Based Signal Enhancement. These techniques not only effectively mitigate various underwater noise components but also significantly enhance signal quality for ultra-long-range underwater reliable communication.
- We implement *HydroChirp* on different commodity mobile devices and extensively evaluate its performance across various scenarios. The evaluation results demonstrate that *HydroChirp* achieves a communication range of up to 150 meters in high-noise environments. Furthermore, *HydroChirp* effectively preserves battery life on commodity devices while maintaining high communication reliability.

II. OBSERVATION AND ANALYSIS

A. Generic and Low-Bandwidth Noise

In underwater communication, ambient acoustic noise can significantly reduce communication reliability and range. Additionally, due to the hardware limitations of commodity mobile devices, their speakers cannot generate extremely loud acoustic signals. Therefore, it is important to analyze noise components in underwater environments and mitigate their effects. During the experiments, two smartphones (Apple iPhone 15) were used to conduct underwater communication. The sender continuously transmitted acoustic chirp signals to the receiver. The bandwidth of each chirp signal was set to 2000Hz . The transmission power was set to 50% of the maximum allowed sound volume. The communication distance was set to 20m , while the depth was set to 3m .

- **Generic Noise.** Figure 2 (a) shows the spectrum of the detected signals by the receiver. As shown in the figure, the detected chirp signal is relatively weak. Moreover, the receiver also detects high noise signals with frequency range from 3300Hz to 4600Hz . These noise signals introduce significant interference to the transmitted chirps, which reduces communication reliability. In this paper, such wideband noise is defined as **Generic Noise**, as it is relatively stable and consistently

appears in similar wide frequency bands. Furthermore, during our experiments, we observed that higher frequency bands have lower levels of generic noise. However, due to the small size of speakers in commodity devices, the high-frequency signals generated by these speakers have inevitable distortion, which cannot be used for underwater communication.

- **Low-Bandwidth Noise.** As shown in Figure 2 (b), the inherent uncertainty of underwater environments also generates noise with relatively low bandwidth. In this paper, we define this noise as **low-bandwidth noise**. In practice, even if the level of the generic noise is low, these low-bandwidth noise signals can still introduce significant frequency offsets and distortions in transmitted chirp signals. As a result, simply selecting a seemingly clear channel with minimal generic noise cannot ensure reliable underwater communication, as the presence of low-bandwidth noise may still disrupt it.

In summary, it is essential to mitigate the effects of generic and low-bandwidth noise for reliable and ultra-long-range underwater communication.

B. Water Fluctuations and Object Noise

We conduct experiments to analyze noise generated by water fluctuations and specific objects in the same environment.

- **Water Fluctuation Noise.** As shown in Figure 2 (c), the energy of water fluctuation noise is extremely high, which introduces very high distortions to the communication channel. Therefore, to clearly illustrate the effects of these noises on the transmitted chirps, we increased the transmission power to 100% of the maximum allowable sound volume. As shown in this figure, unlike generic and low-bandwidth noises, the bandwidth of water fluctuation noise is relatively random and hard to predict. these noises vary in both starting and ending frequencies. As a result, it is hard to find a fixed underwater communication channel to avoid these noises.

- **Noise Generated by Specific Objects.** Figure 2 (d) illustrates the impact of noise generated by specific objects. In this experiment, a volunteer was asked to swim in the lake. Interestingly, as shown in this figure, human swimming activities on the surface unexpectedly introduce significant noise into the underwater environment. These noises exhibit repetitive patterns and introduce high distortions to the transmitted chirps. In addition, they also have the same bandwidth with similar starting and ending frequencies. However, it is still very hard to mitigate the effect of these noises. This is because multiple objects and activities may introduce diverse noise patterns as the communication distance increases. Some of the noise patterns are relatively weak and do not significantly affect communication signals. On the other hand, although some occasional activities or noises may introduce high distortions to a single chirp, addressing these occasional noises can lead to high system overhead, which will reduce the throughput of the underwater communication system.

In summary, it is crucial to build models to mitigate the noise generated by water fluctuations. Meanwhile, we should design an efficient scheme to selectively address the noise generated by specific underwater objects.

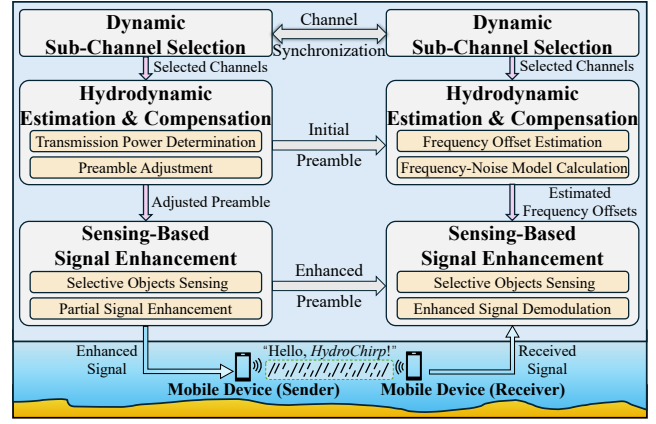


Fig. 3: The Design Overview of *HydroChirp*

III. DESIGN OVERVIEW

The **design goal** of *HydroChirp* is to achieve reliable and ultra-long-range underwater communication using commodity mobile devices. To do this, as discussed in Section I, we need to overcome three key challenges: (i) dynamically selecting channels with relatively low signal distortions; (ii) mitigating interference generated by water fluctuations and adjusting the transmission power; and (iii) effectively sensing and mitigating noise generated by specific underwater objects. Figure 3 shows the overview of *HydroChirp*, which mainly consists of three key components: (i) Dynamic Sub-Channel Selection, (ii) Hydrodynamic Estimation and Compensation, and (iii) Sensing-Based Signal Enhancement and Compensation.

- **Dynamic Sub-Channel Selection (DSS):** The goal of DSS module is to select sub-channels with relatively low signal distortions dynamically. As discussed in Section II, due to the inevitable noise in underwater environments, it is difficult for the receiver to detect and demodulate the acoustic chirp signals transmitted from the sender. To overcome this, our DSS module leverages a novel **channel and chirp segmentation algorithm** to dynamically select low-distortion channels and divide chirp signals to mitigate the effects of noise. Specifically, to avoid the generic underwater noise, the sender and receiver will first synchronize with each other and select the channel with relatively low distortions. Then, to avoid low-bandwidth channel noise and improve segmentation granularity, the receiver dynamically segments the received chirp into multiple chirp segments and drops the segment that is highly affected by the noise. This segmentation process will continue until the remaining chirp segments can be detected with high reliability. Finally, the channels used by the remaining chirp segments are defined as sub-channels and the selection results are synchronized with the sender. In other words, the received chirp is segmented and filtered into multiple nonconsecutive chirp segments, with each segment corresponding to a sub-channel. These high-quality sub-channels will be used for reliable and ultra-long-range communication in noisy underwater environments.

- **Hydrodynamic Estimation & Compensation (HEC):** The HEC module aims to estimate and mitigate severe

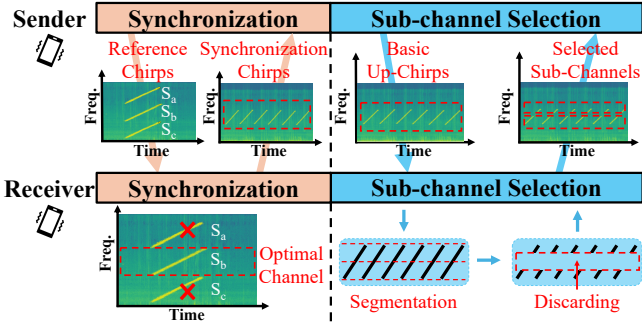


Fig. 4: The Dynamic Sub-Channel Selection module

frequency offsets while dynamically adjusting transmission power in underwater environments. Specifically, to improve communication reliability and range, we consider the water fluctuations generated by surface and internal waves [16], [17]. In real-world scenarios, these fluctuations will not only contribute to frequency offset but also introduce high noise in underwater environments. In our design, instead of simply filtering out the noise, we model the water fluctuation model to not only estimate the frequency offset but also determine the optimal transmission power. Specifically, the speed of the surface and internal waves will introduce extremely high frequency offset to the transmitted signal [20]. Moreover, a high-speed surface or internal wave can generate much higher noise. In *HydroChirp*, the preamble contains multiple calibration chirps for frequency offset estimation. According to the frequency offset, we develop a novel **frequency-noise model** to estimate the noise generated by water fluctuations and determine the optimal transmission power.

- **Sensing-Based Signal Enhancement (SSE):** The SSE module is designed to dynamically sense noises generated by specific objects and perform targeted signal enhancement. Instead of simply filtering out the noise, the **key novelty** of SSE is conducting partial signal enhancement and compensation based on sensing results. Specifically, SSE module consists of two key components: **Selective Objects Sensing** and **Partial Signal Enhancement**. In the Selective Objects Sensing, the sender and receiver collaboratively sense noises generated by specific underwater objects and differentiate their spectral patterns. During the Partial Signal Enhancement, the sender dynamically adjusts the spreading factors for specific chirp segments to enhance the reliability of these noise-affected chirps. Meanwhile, the receiver will leverage a dynamic demodulation strategy to identify and demodulate these chirps.

IV. DETAILED DESIGN

A. Dynamic Sub-Channel Selection

Due to the severe underwater noise, it is extremely hard to detect and demodulate acoustic signals. To overcome this, the goal of Dynamic Sub-Channel Selection (DSS) module is to mitigate the effects of generic and low-bandwidth noises. Figure 4 shows an overview of the DSS module, which consists of two parts: synchronization and sub-channel selection.

- **Synchronization.** During the synchronization procedure, we introduce a handshake scheme to select the optimal

communication channel. Specifically, the sender first transmits multiple pre-defined reference up-chirp signals that have different starting and ending frequencies. To improve transmission efficiency and reduce delay, these up-chirp signals are transmitted at the same time. For example, as shown in Figure 4, the sender transmits three different reference chirps (S_a , S_b , and S_c) simultaneously. The transmitted signal contains mixed frequency components. At the receiver side, it simultaneously demodulates the signals by multiplying them with the corresponding down-chirp signals. Then, it measures the chirp-level Signal-to-Noise Ratio (SNR) for each reference chirp. The channel corresponding to the reference chirp with the highest SNR is selected as the optimal channel, while the remaining channels (S_a and S_c) will be ignored. After that, the receiver transmits multiple synchronization chirps through the selected optimal channel to inform the sender of the selection.

Determining the number of channels for reference chirps transmission n_{rc} is challenging. Specifically, due to the extremely high underwater noise in low-frequency bands and the hardware limitations of mobile device speakers [15], the reference channels are restricted to the range of $2kHz$ to $14kHz$. Meanwhile, for each channel, the bandwidth of each channel should be as small as possible to improve the channel selection granularity and avoid severe channel noise. However, this intuitive approach will significantly reduce the throughput of the communication system. To balance channel selection granularity and system throughput, the maximum allowable bandwidth in our design is set to $2kHz$ and n_{rc} is 6.

- **Sub-Channel Selection.** The selection of the optimal channel can efficiently avoid the generic noise for ultra-long-range underwater communication. However, as introduced in Section II-A, the low-bandwidth noise may also affect communication reliability. To overcome this challenge, we propose a novel **channel and chirp segmentation algorithm** to divide the chirp and the corresponding channel into smaller segments. As shown in Figure 4, during the sub-channel selection, the sender transmits several basic up-chirps through the selected optimal channel. On the receiver side, the optimal channel is divided into multiple channel segments. The receiver then demodulates the basic up-chirps within each channel segment by multiplying corresponding down-chirps. If certain chirps cannot be distinguished in the frequency domain, it indicates that they are heavily influenced by low-bandwidth noise, and the corresponding channel segments should be discarded.

Ideally, the number of corresponding channel segments should be as high as possible to accurately identify the noise-affected channel segments. However, this process will significantly increase the system overhead and reduce the overall processing speed. On the contrary, if the number of channel segments is low, some partially affected channel segments may be discarded, which will reduce the available communication spectrum and impact the system throughput. To solve this, in our design, the number of segments is determined based on the bandwidth of the noise. Specifically, as discussed in Section II-A, the bandwidth of low-bandwidth noise is around $100Hz$. Then, to better preserve the spectrum, the bandwidth of each

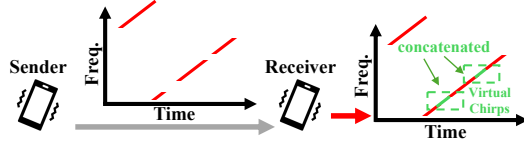


Fig. 5: An Example of Concatenating Nonconsecutive Chirp With Virtually Generated Chirp Segments

channel segment should be set to at most half of the bandwidth of the noise. Since the bandwidth of low-bandwidth noise is approximately 100Hz , each channel segment is set to 50Hz . In other words, a 2kHz optimal channel should be divided into 40 segments to accurately identify the noise-affected channel segments while maintaining processing efficiency. At last, the channels used by these nonconsecutive chirp segments are defined as sub-channels. *HydroChirp* will use these sub-channels to transmit chirp signals to the receiver. In practice, the receiver will keep calculating the current bit error rate. If it is lower than a predefined threshold, it will inform the sender to initialize this process again.

We want to mention that using nonconsecutive acoustic chirps does not affect the embedded information. The sender can still use traditional Chirp Spread Spectrum (CSS) modulation schemes [13] to encode bits into the acoustic signals. As illustrated in Figure 5, the sender transmits a modulated chirp to the receiver. Unlike traditional consecutive chirps, the transmitted chirp is nonconsecutive to avoid generic and low-bandwidth noises. On the receiver side, the nonconsecutive chirp segments are concatenated with virtual chirp segments. **To clarify**, these virtual chirp segments are generated using the same spreading factor (SF) as the nonconsecutive chirp segments, while their frequency ranges correspond to that of generic and low-bandwidth noises. After concatenation, the received signal can be reconstructed into a **complete chirp signal without interference**. The receiver can demodulate the signal by multiplying the corresponding down-chirp signals.

B. HydroDynamic Estimation & Compensation

In real-world scenarios, water fluctuations caused by surface and internal waves will introduce significant uncertainty to the underwater communication system, including large frequency offsets and high levels of noise [16]. Moreover, as the communication distance increases, the number of waves which can affect the transmitted acoustic signals also increases, which further affects communication reliability. In addition, different water environments exhibit distinct channel characteristics, such as varying noise levels and frequency offsets. To overcome these, *HydroChirp* utilizes a HydroDynamic Estimation & Compensation (HEC) module to estimate the frequency offsets and noises introduced by water fluctuations. Specifically, the HEC module consists of three steps: (i) initial preamble transmission and frequency offset estimation; (ii) frequency-noise model calculation; and (iii) optimal transmission power determination and preamble adjustment.

• **Initial Preamble Transmission and Frequency Offset Estimation.** In *HydroChirp*, we use the short preamble of the packet transmitted by the sender to estimate the frequency

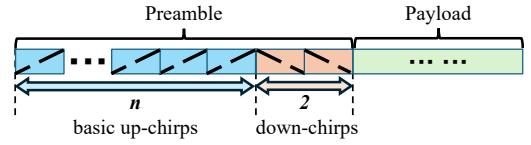


Fig. 6: The Packet Structure of *HydroChirp*

offset and reduce the system overhead. As shown in Figure 6, the preamble consists of n basic up-chirps and two down-chirps and n can be dynamically changed according to the water environments. During the initial preamble transmission and frequency offset estimation step, n is first set to 4, while the two down-chirps are used to inform the receiver of the starting position of the data packet. After receiving the data packet, the receiver will compute the frequency offset Δf using $\Delta f = f_d - f_{start}$, where f_d is the detected starting frequency and f_{start} is the ideal frequency of the chirp.

However, unlike communication systems that transmit packets through the air, in underwater communication, the frequency offset Δf calculated from the preamble cannot be directly applied to compensate for the chirps in the payload. This is because the inertia of surface and internal waves will introduce inevitable frequency offset correlations between the adjacent chirps. To address this, we design a self-regressive model to accurately estimate the frequency offset of the current chirp based on the frequency offset of the previous chirp. Specifically, the frequency offset of the current chirp, Δf_i , can be estimated as: $\Delta f_i = \phi \Delta f_{i-1}$, where Δf_{i-1} is the frequency offset of the previous chirp. ϕ is the regression coefficient, which can be calculated using:

$$\phi = \frac{\sum_{i=1}^n (\Delta f_i - \bar{\Delta f})(\Delta f_{i-1} - \bar{\Delta f})}{\sum_{i=1}^n (\Delta f_i - \bar{\Delta f})^2} \quad (1)$$

Here, Δf_i is the frequency offset of i -th chirp in the payload and $\bar{\Delta f}$ is the average frequency offset in the preamble.

• **Frequency-Noise Model Calculation.** To further improve communication distance and reliability, we introduce an innovative frequency-noise model that mitigates noise caused by surface and internal waves while determining the optimal transmission power. Specifically, based on the Doppler Shifts formula [21], the short-term water wave velocity v_i for a given chirp i can be estimated as $v_i = \frac{\Delta f_i}{f_i^0} c$, where Δf_i is the frequency offset of chirp i and f_i^0 is the start frequency of the chirp. c is the speed of sound in water and can be set to 1500m/s [22]. According to the research in underwater wave-generated noise estimation [23], [24], we propose a novel hydrodynamics model to estimate the short-term noise level l using the frequency offset:

$$l \approx C \omega^{-5} e^{(-\frac{5}{4}(\frac{\omega_p}{\omega})^4)} \quad (2)$$

Here, l denotes the spectral density of the noise. C is an empirical parameter with a value of 4.8×10^4 [23], [24]. The term ω_p represents the peak frequency of the wave spectrum. The exponential term, $e^{(-\frac{5}{4}(\frac{\omega_p}{\omega})^4)}$, represents the low-frequency cutoff of the wave spectrum. The angular frequency ω for a specific chirp i is calculated as $\omega = 2\pi f_i$.

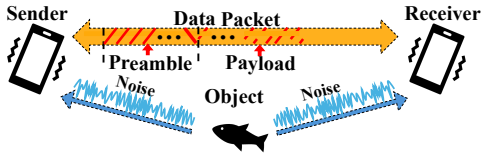


Fig. 7: An Overview of the Selective Object Sensing

By using Equation 2, the receiver will obtain a noise level array $L = (l_1, \dots, l_n)$, where l_i is the estimated short-time wave noise level of i -th chirp. However, since L is a non-monotonic array, it is challenging to directly get the noise level trend. To address this, we utilize the average changing rate of the noise level T for transmission power adjustment. T can be calculated as $T = \frac{1}{n-1} \sum_{i=1}^{n-1} (\frac{l_{i+1}-l_i}{l_i})$, where n is the number of chirps and l_i is the i -th element in L .

• **Optimal Transmission Power Determination and Preamble Adjustment.** At last, the receiver transmits the calculated T back to the sender. In practice, the amplitude and strength of water waves vary depending on the application scenarios (e.g., swimming pool, lake, river, etc.), which limits the range of T to values between -1 and x , where x is scenario-dependent. Meanwhile, due to hardware limitations, the sound volume on commodity mobile devices (e.g., smartphones) can only be adjusted in fixed increments. To minimize system overhead, our design restricts transmission power to five discrete levels: 5%, 25%, 50%, 75%, and 100% of the maximum volume. These levels are mapped to the range of T . For example, if the range of T is $[-1, 1]$ and the current T is 0, the optimal transmission power is set to 50% of the maximum sound volume. Meanwhile, if the current T is -1, we can select the minimal sound volume 5% to preserve the battery life.

It is also important to note that a very high value of T typically indicates stronger water fluctuations and higher surface and internal wave speeds, which will introduce very high uncertainty to the frequency offset. In the initial preamble transmission and frequency offset estimation step, the default number of up-chirps, n , is set to 4. However, this may be insufficient to adapt to high water fluctuations. To improve communication reliability, the number of up-chirps in the preamble will also be dynamically adjusted based on the value of T to accurately estimate the frequency offset.

C. Sensing-Based Signal Enhancement

In underwater scenarios, the noise generated by specific objects (e.g., paddling activities, fountain, fish, etc.) can cause significant interference in certain sub-channels. This interference will eventually introduce severe distortions to the corresponding nonconsecutive chirps, which reduces communication reliability. To address this issue, the Sensing-Based Signal Enhancement (SSE) module introduces a novel approach to dynamically sense object-generated noise and enhance the reliability of targeted chirps. Specifically, our SSE module comprises two key components: **Selective Objects Sensing** and **Partial Signal Enhancement**.

• **Selective Object Sensing.** In our design, the preamble in Figure 6 is also used to sense noise generated by specific

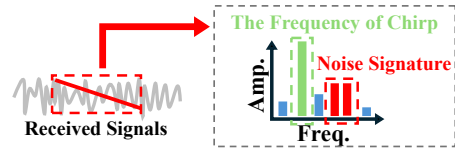


Fig. 8: Our Noise Signature Recognition Process

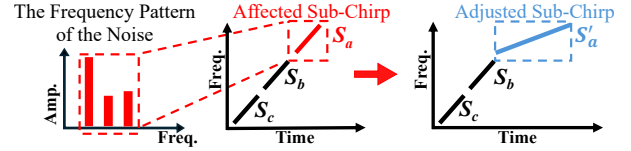


Fig. 9: Dynamic SF Adjustment on the Affected Chirp

objects. As shown in Figure 7, the sender and receiver use the up-chirps in the preamble to estimate the frequency offset and noise generated by water fluctuations. Upon detecting the preamble chirps, the receiver will analyze the distribution of the FFT bins during the demodulation process. In Figure 8, the highest bin represents the frequency of the demodulated chirp signal. However, since this signal is mixed with the noise generated by specific objects, it is hard to estimate the noise pattern based on this single bin. To address this, we utilize the fact that noise signals consist of multiple frequency components. In our design, the first two highest noise bins are considered as the noise signature. The receiver will record these signatures and their occurrence count with a hash table as the dataset, where the value of each signature is the indexes of two highest noise bin. If a signature is detected multiple times within a short time window and matches a known signature in the dataset, it indicates that this specific object may continually affect the underwater communication. In this case, *HydroChirp* will adjust the chirp configuration to enhance the reliability of the affected chirps. On the other hand, if a frequent signature is not found in the dataset during the communication, the receiver updates its dataset to adapt future communication. We also want to mention that this design maintains low time and space overhead. By using hash table, the time complexity of signature matching is $O(n)$. Meanwhile, as the index of a noise bin requires at most 4 bits, the dataset remains at the kilobyte level.

• **Partial Signal Enhancement.** To conduct chirps adjustments, the sender modifies the chirps in the affected sub-channel based on the identified noise signatures. Specifically, as shown in Figure 9, only the sub-chirp S_a within the affected sub-channel is altered by the noise frequency pattern. Then, the sender performs dynamic spreading factor (SF) adjustment to modify S_a into an adjusted sub-chirp S'_a . The key advantage of this approach is that it enhances throughput by limiting adjustments to the small affected sub-chirp, rather than the entire chirp. This minimizes unnecessary overhead and preserves the efficiency of overall communication system.

To do this, the sender incrementally increases the SF by a step length of 1. For each SF, the sender will generate the adjusted sub-chirp S'_a in the preamble of the packet. This process continues until all basic up-chirps in the preamble are adjusted. For example, if S_a is initially generated using

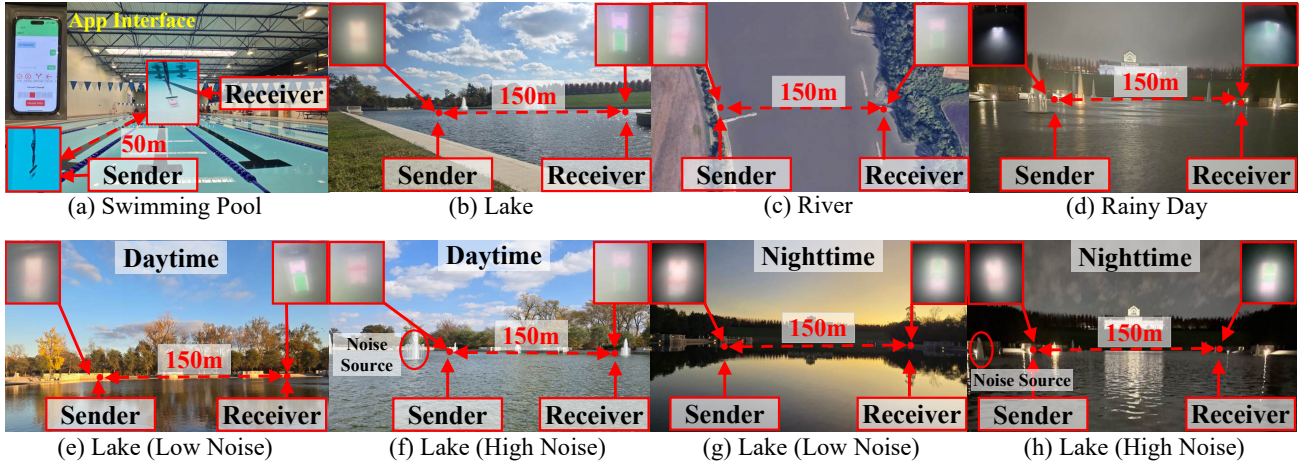


Fig. 10: Our experiment scenarios include: (a) swimming pool, (b) lake, (c) river, (d) rainy day, (e) daytime lake with low noise, (f) daytime lake with high noise, (g) nighttime lake with low noise, and (h) nighttime with high noise.

SF = 7, the sender will generate S'_a with SF values of 8, 9, 10, 11, and 12. This strategy supports a wide range of chirp configurations and accommodates preambles of different lengths. After chirp adjustments, the sender transmits the modified signals to the receiver. The receiver first identifies the affected sub-channel based on the noise signatures and applies the same SF adjustments to the down-chirps. It then demodulates the received signals using the adjusted down-chirps. After that, the receiver calculates the Signal-to-Noise Ratio (SNR) for each adjusted chirp using $SNR = \frac{P_s}{P_n}$, where P_s is the chirp's signal power and P_n is the average noise power. This SNR value will be used to determine the minimum SF value required for successful demodulation. At last, the receiver transmits this SF value back to the sender. After receiving the updated SF value, the sender will adopt the optimized chirp configuration for future communication.

V. IMPLEMENTATION AND EVALUATION

A. Implementation

We implement *HydroChirp* on various commodity mobile devices, including the Apple iPhone 13, Apple iPhone 14 Pro Max, Apple iPhone 15, and Samsung Galaxy S24+. The operating systems are iOS 17 for the iPhones and Android 14 for the Samsung Galaxy. Although current smart devices can support IP68 waterproof standards, due to the uncertainty in outdoor underwater scenarios, we use additional waterproof cases to protect the mobile devices. These cases are cost-effective, priced at only \$8.99 for a set of two [25]. Since the experiment results of all these devices show a similar trend, we show the experiment results of the Apple iPhone 15.

Figure 10 (a) shows the *HydroChirp* application interface, where users can read, type, and select pre-defined messages for transmission to the receiver. During the experiments, the initial preamble of a packet consists of 4 basic up-chirps and 2 down-chirps, which is dynamically adjusted by the HEC module. In addition, the SSE module contains pre-defined noise signatures, including fountain and paddling activities.

As shown in Figure 10, we extensively evaluate *HydroChirp* across various scenarios, times, and weather conditions, including an indoor swimming pool, a lake, a river, and rainy days during both daytime and nighttime. These scenarios have different noise levels, underwater objects, and water fluctuation patterns. The length of our indoor swimming pool is 50m. During our experiments, around 5 to 8 students were swimming in it. In contrast, the lake contains many active fountains, which generate much higher noise during experiments. Meanwhile, the river exhibits significant water fluctuations with more than 200,000 Cubic Feet per Second and high surface and internal flow speeds [26].

Since this is the first work investigating reliable and ultra-long-range underwater communication using commodity mobile devices, the state-of-the-art research is complementary but does not provide appropriate baselines for comparison. To show the advantages of *HydroChirp*, we use the **Chirp Spread Spectrum (CSS) scheme** [13] as our baseline. In this approach, the sender directly transmits encoded acoustic chirp signals to the receiver without sensing the channel or noise. Because the CSS-based approach cannot dynamically adjust transmission power, we set the sound volume to 100% of the maximum allowed sound volume of the mobile devices during the experiment. To further show the benefits of our design, we also implement two basic *HydroChirp* approaches: **HydroChirp-DSS** and **HydroChirp-HEC**.

- **HydroChirp-DSS**: This approach does not use the Dynamic Sub-Channel Selection (DSS) module to avoid generic and low-bandwidth noises in underwater scenarios. Additionally, the Sensing-Based Signal Enhancement (SSE) module is also disabled as it requires chirp signal adjustments at the sub-channel level. Consequently, this approach randomly selects a channel for underwater communication and relies on the HydroDynamic Estimation & Compensation (HEC) module to mitigate noise generated by water fluctuations.

- **HydroChirp-HEC**: This approach does not use the HydroDynamic Estimation & Compensation module to compensate for the signal affected by water fluctuations. However,

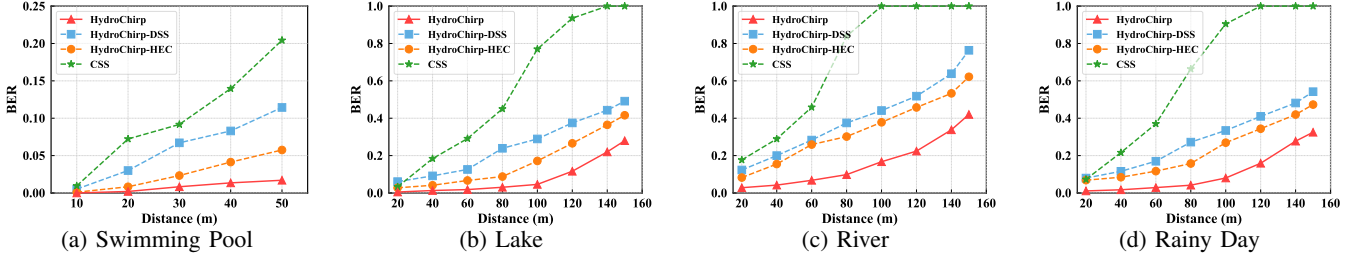


Fig. 11: The BER of *HydroChirp* in Different Scenarios

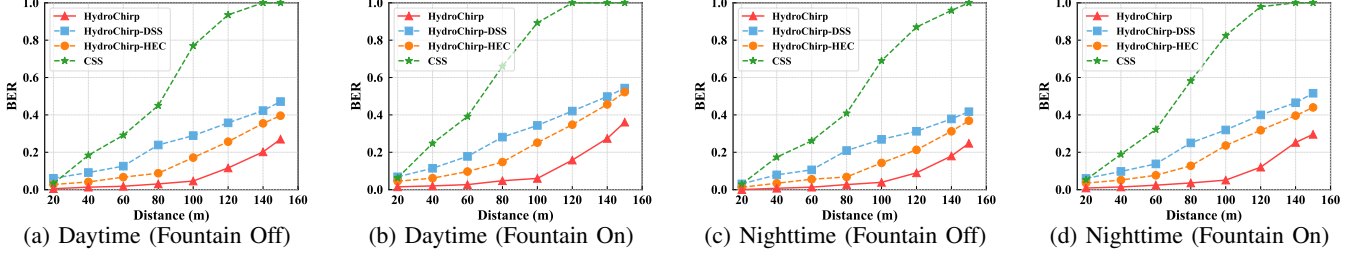


Fig. 12: The BER of *HydroChirp* in a Lake under Different Noise Levels and Times

since it can leverage the Dynamic Sub-Channel Selection (DSS) module to select optimal channels and generate nonconsecutive chirps, the Sensing-Based Signal Enhancement (SSE) module remains active in this approach.

B. Evaluation Results

1) **Bit Error Rate:** In Figure 11, we show the Bit Error Rate (BER) of *HydroChirp* under various scenarios and conditions, including an indoor swimming pool, lake, river, and rainy day.

• **Indoor Scenarios: Swimming Pool.** As shown in Figure 11 (a), the BER of *HydroChirp* is significantly lower than that of state-of-the-art solutions. The CSS-based approach exhibits the highest BER, since it cannot select the optimal communication channel and mitigate the noise from water fluctuations. When the communication distance reaches 50m, the BER of CSS is approximately 0.21. Meanwhile, *HydroChirp*-DSS performs significantly worse than *HydroChirp*-HEC. This is because *HydroChirp*-DSS can only mitigate the noise from water fluctuations. However, with multiple students swimming in the pool, it is crucial to select the optimal channel and detect the repetitive noise generated by swimming activities. As a result, at a distance of 50m, the BER of *HydroChirp*-HEC is around 0.056, while *HydroChirp*-DSS records a BER of 0.12.

The performance of *HydroChirp* is the best. At a communication distance of 50m, the BER is less than 0.02, which is **10.5 times**, **6 times**, and **2.8 times** lower than that of CSS, *HydroChirp*-DSS, and *HydroChirp*-HEC, respectively. This is because *HydroChirp* effectively selects the optimal channel and mitigates noise generated by water fluctuations. Additionally, it is important to emphasize that *HydroChirp* does not rely on pre-recorded noise signatures of swimming activities. During our experiments, *HydroChirp* dynamically recorded these repetitive noise signatures and adjusted the chirp signals accordingly. **In summary**, *HydroChirp* can effectively support reliable underwater communication in indoor scenarios.

• **Outdoor Scenarios: Lake and River.** As shown in Figure 11 (b) and Figure 11 (c), the performance of *HydroChirp* in lake and river shows similar trends. Specifically, the *HydroChirp* is able to achieve ultra-long-range underwater communication in both scenarios. When the communication distance reaches 150m, the corresponding BERs of *HydroChirp* is around 0.265. It is important to note that the river is noisy with more than 200,000 Cubic Feet per Second [26]. However, even in this worst scenario, *HydroChirp* is still able to conduct communication with around 0.41 bit error rate. This is because *HydroChirp* can dynamically select optimal communication channels, estimate the water fluctuation noise, and adjust the sub-channel chirps to mitigate the randomness and noise in underwater environments. On the contrary, the CSS-based approach cannot support long-range underwater communication. When the communication distance reaches 100m, the BERs are 0.79 and 1 for lake and river scenarios, respectively. **In summary**, *HydroChirp* can support reliable and ultra-long-range communication distance in severe underwater environments.

• **Rainy Day.** In Figure 11 (d), we conduct experiments in the lake scenario during a rainy day. This experiment is particularly challenging due to the highly random noise generated by raindrops. The performance of *HydroChirp* remains significantly better than state-of-the-art solutions. Compared to Figure 11 (b), the raindrops have almost no effect on *HydroChirp*. However, for the CSS-based approach, communication reliability is drastically reduced. The BER of CSS-based approach reaches 1 when the communication distance is 120m. **In summary**, *HydroChirp* demonstrates robust performance in severe weather conditions.

2) **Different Noise Levels and Times:** We evaluate the performance of *HydroChirp* under different noise conditions and times in the lake scenario, as shown in Figures 11 (e) to (h), where fountains serve as the primary noise source.

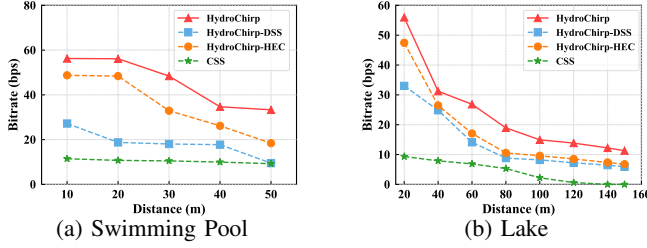


Fig. 13: The Throughput of *HydroChirp*

• **Daytime.** Figures 12 (a) and 12 (b) present the BERs during daytime. Specifically, the BERs of *HydroChirp* remain extremely low when the communication distance is below 100m, regardless of the noise conditions. In contrast, the BERs of state-of-the-art solutions increase rapidly when the fountains are active. This is due to the high underwater noise and water fluctuations generated by the multiple active fountains in the lake. As shown in Figure 12 (b), when the communication distance reaches 100m, the performance of *HydroChirp* is around **18 times**, **5.6 times**, and **4.7 times** better than CSS, HydroChirp-DSS, and HydroChirp-HEC, respectively. When the distance is 150m, the BER of *HydroChirp* is still lower than 0.4. Meanwhile, the BERs of CSS, HydroChirp-DSS, and HydroChirp-HEC are 1, 0.57, and 0.56, respectively.

• **Nighttime.** Figures 12 (c) and 12 (d) show the performance of *HydroChirp* during nighttime. All the approaches perform slightly better, as the lake is quieter at night with relatively stable surface waves. In this experiment, *HydroChirp* continues to outperform the other approaches, as it effectively selects the optimal underwater communication channel and estimates water fluctuations based on the frequency offsets introduced by the fountains. **In summary**, *HydroChirp* can support reliable and ultra-long-range communication across different times of day, even in environments where water fluctuations are caused by artificial sources such as fountains.

3) **Throughput:** Figure 13 shows the throughput of *HydroChirp*. In the swimming pool scenario (Figure 13 (a)), the throughput of *HydroChirp* is very stable (around 34 bps), regardless of communication distance. Meanwhile, in the outdoor lake scenario (Figure 13 (b)), the throughput of *HydroChirp* reduces as the communication distance increases. When the communication distance reaches 150m, the throughput of *HydroChirp* is still as high as 11bps. We believe that this throughput is high enough as the messages transmitted in underwater scenarios are normally relatively short (e.g., "out of air," "go left," etc.). We want to mention that during the experiments, the fountain in the lake is active and keep generating high noise. Moreover, compared to the throughput reported in AquaApp [7] (around 10 - 20 bps) at 100m, *HydroChirp* can achieve much longer communication distance (150m) with similar throughput in extremely noisy scenarios.

VI. RELATED WORK

Optical-Signal-Based Solution. Researchers have demonstrated that optical signals have the potential to support un-

derwater communication [28]–[31]. These approaches mainly utilize LEDs to generate visible light signals for underwater communication and the communication distances are up to 50m [28]. For example, Shrimp [29] utilizes customized LEDs to achieve reliable underwater communication over distances up to 3m. Meanwhile, researchers have also explored using complex deep learning models to demodulate underwater light signals [31]. To further improve the performance of light-signal-based underwater communication, channel models for underwater communication have been proposed to improve communication reliability and distance [30].

Acoustic-Signal-Based Solution. To improve communication distance, researchers also investigate acoustic-signal-based solutions [7], [32]–[34]. Most of them require customized sensors to transmit and detect acoustic signals [33]. Recently, AquaHelper [12] introduced a novel system utilizing multiple receivers to detect acoustic SOS messages in swimming pools. Building on this, NeuSOS [34] employs a deep learning model to enhance communication reliability in indoor swimming pool scenarios. The most relevant work is AquaApp [7], which focuses on high-throughput underwater communication using OFDM acoustic signals generated by mobile devices.

Although existing approaches perform well in their application scenarios, they require customized hardware for short-distance underwater communication. In contrast, we focus on ultra-long-range reliable and underwater communication using commodity mobile devices. Moreover, *HydroChirp* can dynamically select the optimal channel, senses and models water fluctuations, and address noise generated by specific underwater objects to improve underwater communication reliability and distance at the same time.

VII. CONCLUSION

In this paper, we present *HydroChirp*, the first acoustic underwater communication system for reliable and ultra-long-range communication in diverse scenarios using commodity mobile devices. Specifically, to improve communication reliability and mitigate inevitable underwater noise, we introduce a novel Dynamic Sub-Channel Selection (DSS) module that dynamically selects low-distortion channels and segments chirp signals to minimize noise effects. To address frequency offsets and extend communication range, we leverage water fluctuation patterns for signal compensation and transmission power adjustment. Additionally, we are the first to address noise generated by underwater objects and improve the reliability of specific chirps. Extensive experiments show that *HydroChirp* can achieve a communication range of up to 150 meters with high reliability. We believe that *HydroChirp* will open a new door for future underwater communication techniques.

VIII. ACKNOWLEDGEMENT

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